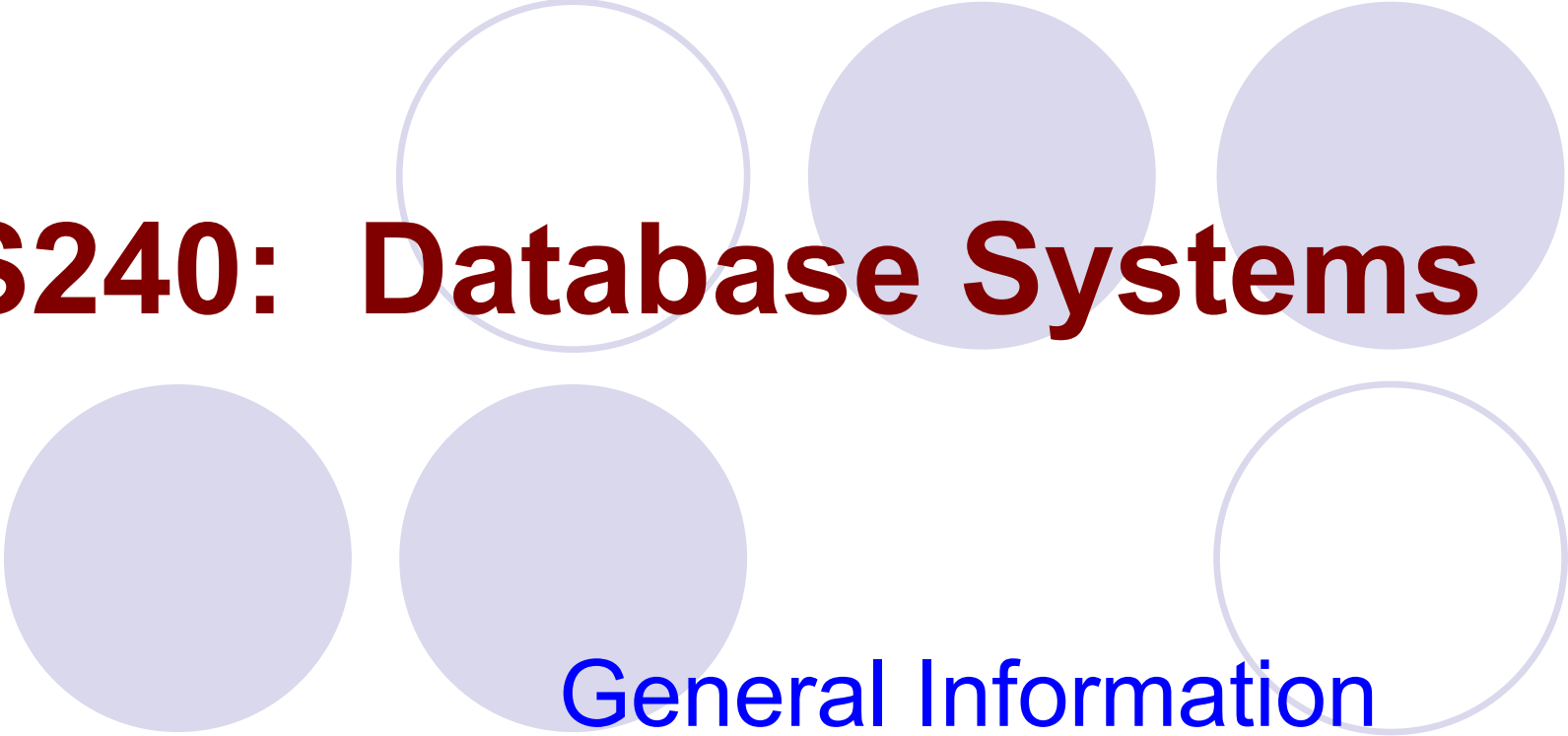




CS240: Database Systems

General Information

People

Instructor:

- Dr. Yuan-Yuan Pan 潘媛媛
- Email: yypan@must.edu.mo
- Room: A306a

Office Hours:

- TUE 12:00 – 14:00
- WED 14:30 – 17:30
- THU 10:00 – 12:00
- FRI 12:00 – 15:00

Course Description

- The objective of the course is to study the concepts and principles of database management systems (DBMS).
- The first part of the course deals with the basic concepts, the general system structure, data modeling, database languages, SQL, relational database design principles, and schema refinement and normal forms.

Course Description

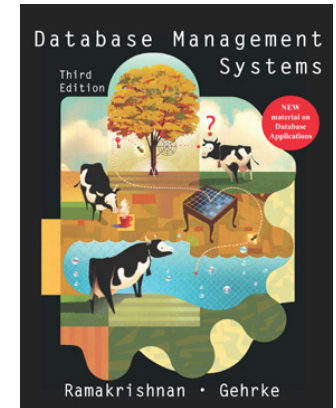
- The second part deals with file system organizations, indexing methods, query optimization, recovery mechanisms, concurrency control, and transaction management.

Course Outcomes

- Upon successful completion of this course, students will be able to
 - Understand the basic concepts and architecture of database system.
 - Master writing queries in relational data languages including SQL and relational algebra.
 - Master transaction and concurrency control
 - Be competent with conceptual database design.
 - Be competent with logical database design.
 - Be competent with schema refinement.

Textbook

- *Database Management Systems*, by Raghu Ramakrishnan, Johannes Gehrke, McGraw Hill (3rd edition), 2003.



- 數據庫管理系統原理與設計(第3版), 周立柱 張志強 李超 王煜譯, 清華大學出版社

Reference Books

- *Database System Concepts, 7th Edition.*
Abraham Silberschatz, Henry F. Korth, S. Sudarshan.
McGraw-Hill Education 2019.
- *Fundamentals of Database Systems, 7th Edition.*
Ramez Elmasri, Shamkant B. Navathe.
Pearson, Prentice Hall, 2017.
- 數據庫系統概論（第五版），王珊 薩師煊編著

Course Outline

1. Introduction to Database systems
2. E/R Models
3. The Relational Model
4. Relational Algebra and Calculus
5. SQL
6. Schema Refinement and Normal Forms
7. Tree-Structured Indexing
8. Hash-based Indexing
9. Overview of Query Evaluation
10. Overview of Transaction Management

Assessment Policy

- Class attendance 10%
- Assignments 20%
- Quiz 20%
- Final Exam 50%

Assignments

No late submission!

Number of Late Days	Deduction (out of 100 marks)
1	-10
2	-20
3	-30
4	-40
5	-50
5 above	0

In-class exercises and class participation

- There are some in-class exercises.
- The performance and the participation will be counted in the final grade.

Notes

- **No Plagiarism**

- Independent - Don't **copy** from others
- Don't show your answer to other students
- Students violate the rules will get 0 score.